

8-Bit Signed Binary: Sign-Magnitude Representation

Digital Logic Project Report

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1. Introduction

This project demonstrates an 8-bit signed binary system using Sign-Magnitude representation. The design is implemented and simulated in Logisim to connect binary representation theory with practical digital logic design.

2. Problem Statement

Standard unsigned binary representation cannot represent negative numbers because every bit pattern maps to a non-negative value. Sign-Magnitude solves this by using the MSB as a sign flag and the remaining seven bits as the magnitude.

3. Sign-Magnitude Representation

For an 8-bit value, B7 is the sign bit. B7 = 0 represents a positive value and B7 = 1 represents a negative value. B6 through B0 represent the magnitude. The range is -127 to +127, with two representations of zero: +0 and -0.

4. Design and Methodology

The input is divided into a 1-bit sign field and a 7-bit magnitude field. The sign is detected from the MSB, while the lower seven bits are processed independently as the magnitude. The circuit is validated using positive, negative, zero, and boundary cases.

5. Logisim Implementation

The supplied Logisim circuit contains an 8-bit input, a splitter, a sign output for B7, and a 7-bit magnitude output for B6-B0. This provides a clear demonstration of the fundamental sign-magnitude decoding operation.

6. Test Cases

Input	Sign	Magnitude	Expected
00000000	+	0000000	+0
10000000	-	0000000	-0
00000001	+	0000001	+1
10000001	-	0000001	-1
00000101	+	0000101	+5
10000101	-	0000101	-5
01111111	+	1111111	+127
11111111	-	1111111	-127

7. Example

For 10000101, the MSB is 1, so the number is negative. The remaining seven bits are 0000101, which represent magnitude 5. Therefore, the Sign-Magnitude value is -5.

8. Advantages

Sign-Magnitude is intuitive, provides direct sign detection through the MSB, offers a symmetric range, and makes the magnitude directly readable from the lower seven bits.

9. Limitations

The representation has two zero values, requires more complex arithmetic handling for addition and subtraction, and is less commonly used in modern integer hardware than Two's Complement.

10. Conclusion

The project demonstrates the implementation and validation of 8-bit Sign-Magnitude representation in Logisim. It applies binary representation concepts to a practical digital logic circuit and documents the resulting design and test cases.